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S.C. Dwivedi

Build A Real-Time **USB Data Logging System**

A. Robson Benjamin

Presented here is a USB-based real-time data logging system using PIC18F452 microcontroller (MCU), VNC1L-1A USB host controller chip, DS1302 RTC chip and LCD. The system has the flexibility to set time and duration of data measurement.

Data loggers are essential for remote data acquisition systems and generally require storage media to store the acquired data. Storage media, like pen drives, require a device with USB host capability or PC to transfer acquired data. A general-purpose MCU does not have USB host capability. Here is a new data acquisition system with Vinculum chip that provides USB host functionality. Also, by using DS1302 chip and 4x4 keypad, you can set the time and duration of data measurements.

The block diagram of the real-time USB data logger is shown in Fig. 1. The author's prototype board is shown in Fig. 2.

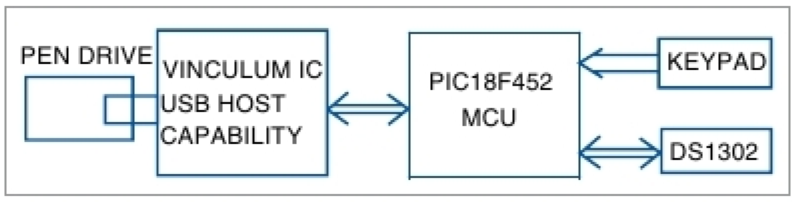

Fig. 1: Block diagram of the real-time USB data logger

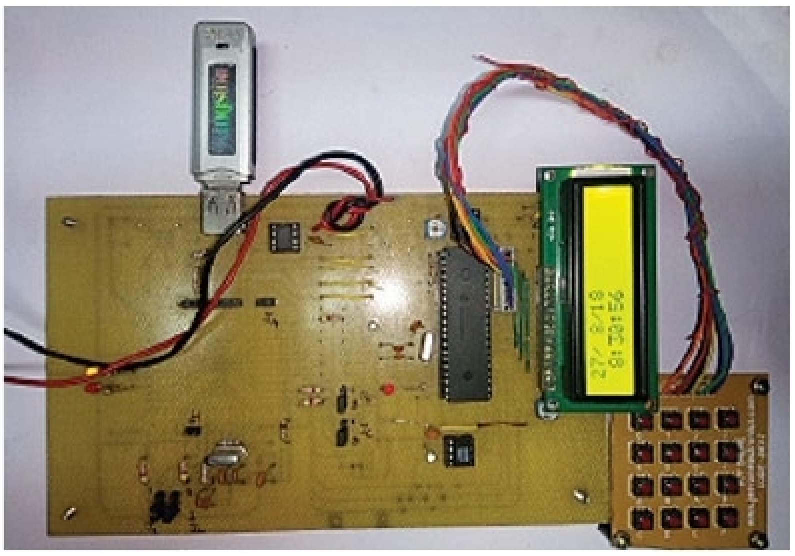

Fig. 2: Author's prototype

Circuit and working

The circuit diagram of the real-time USB data logger is shown in Fig. 3. It is built around 3.3V voltage regulator MAX882 (IC1), temperature sensor LM35 (IC2), Vinculum host controllers VNC1L-1A (IC3) and PIC18F452 (IC4), real-time, trickle charge time-keeping DS1302 chip (IC5), 16x2 liquid crystal display (LCD1) and a few other components.

PIC18F452 MCU provides data acquisition functionality. VNCIL-1A is interfaced to PIC18F452 through UART interface.

VNC1L-1A. It has two USB ports and a combined serial UART, serial peripheral interface (SPI) or FIFO interface. Pins USB1DP and USB1DM carry USB data signals D+ and D- of USB Port-1, respectively. Pins USB2DP and USB2DM carry USB data signals D+ and D- of USB Port-2, respectively.

Port-1 supports USB slave peripherals (not used). Port-2 supports BOMS (bulk only mass storage) class devices such as USB flash drives. Serial UART, SPI or FIFO interface for communication between VNC1L-1A and PIC18F452 can be selected using a pair of pins, ACBUS5 (pin 46) and ACBUS6 (pin 47), by connecting the pins to ground or 3.3V.

UART interface is selected by connecting pins 46 and 47 of IC3 to ground via 47kΩ resistor. ADBUS0 (pin 31) and ADBUS1 (pin 32) carry TxD and RxD signals, and ADBUS2 (pin 33) and ADBUS3 (pin 34) carry RTS# and CTS# signals of UART interface, respectively.

Vinculum provides several pre-compiled standard firmware versions to support different functionalities for VNC1L-1A. For the present application, firmware version VDAP (disk and peripherals) is used. The firmware runs from on-chip 64kB flash memory in VNC1A-1A, and includes a command monitor that allows the external PIC MCU to send and receive command/data via the command monitor interface.

The monitor supports DOS-like disk management commands to perform directory operations (make, delete or list directories), file operations (create, open, read, write, close, delete or rename files) and debug operations (identify or get firmware versions). It also includes commands that allow detection of attachment and removal of flash memory drive to VNC1L-1A.

When the firmware detects a flash drive attached to the USB port, it sends a DOS-like prompt D:\> on the UART interface. It generates codes to flash an LED (LED2) connected to pin 18 when data is written into the flash drive.

The command monitor operates either in command or data mode. In command mode, VNC1L-1A interprets the codes from the monitor port as commands and acts on it. In data mode, it passes data from the monitor port to the USB device. Initially, when VNC1L-1A starts, it stays in command mode. It then signals VNC1L-1A pins ADBUS5 (pin 36) and ADBUS6 (pin 37), which carry DTR# and DSR# signals of serial UART interface, and switches the mode of operation. In the present application, VNC1L-1A is made to operate in command mode by connecting the aforementioned pins to 3.3V through 47kΩ.

Pins ADBUS0 through ADBUS3, PROG#, RESET#, Vcc and GND of VNC1L-1A are used for programming via FTDI programmer.

Remove jumper links at connectors CON4 and CON5 during programming, otherwise pins 9 and 10 should be connected to 3.3V.

More details on Vinculum host controller including FTDI programming are available along with the source code, at *source.efymag.com*